Green Plates and Red Flags: Effect of Public Shaming with License Plates on Tax Evasion

Yongtae Kwon*

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Abstract

Detecting and enforcing tax evasion through the misuse of corporate tax benefits is difficult and prohibitively costly for tax authorities. I study South Korea's recent license plate reform that mandates a distinctive, green license plate for corporate registered cars above a price threshold, introduced to deter individuals purchasing luxury vehicles under corporate registration. I interpret this policy through a social signaling model where the colour and visibility of license plates signal the likelihood and degree of tax evasion. By introducing a visible marker tied to a price threshold, the reform increases both the observability and informational content of this signal. To evaluate the policy, I estimate an event-study design that encompasses the universe of new vehicle registrations. Sales of new corporate-registered cars above the threshold decline immediately after the reform, suggesting an initial deterrence effect. However, sales gradually revert to pre-policy levels within 8 to 10 months. Accounting for anticipation effects, the policy reduced 9,401 luxury corporate vehicles through the first eight months, with limited effect afterwards. These findings suggest that policies relying on socially interpreted signals with uncertain informational content may have limited effectiveness in reducing tax evasion.

Keywords: Tax Evasion, Action Signaling, Public Disclosure, License Plate

JEL Codes: D91, H26, K34, L51

*Second year PhD Student in Public Policy. Lyndon B. Johnson School of Public Affairs, University of Texas at Austin (email:yongtae.kwon@utexas.edu).

1 Introduction

Tax evasion is a persistent problem for many tax authorities around the world. Among the $4,635 billion of total liable taxes in tax year 2022, $606 billion has not been collected in the United States due to underreporting, under-payment or non-filing (Internal Revenue Service, 2024). A fundamental problem is that tax agencies have a limited number of tools to enforce tax compliance. While audits and inspections have been the cornerstone of tax enforcement, it is costly for agencies to increase the probability of audits and thus audit rates have been historically low. For example, less than 1% of firms in the United States have been audited in 2022 (Internal Revenue Service, 2023).

Several policies other than formal audits and inspections have been effective in increasing overall compliance, but their potential for widespread use remains limited. Primarily, third party audits increase the reporting and compliance rate of taxpayers by reducing the possibility of false reporting going unnoticed (Slemrod, 2019). Third party audits, however, increases the compliance burden of taxpayers and may contradict with the purpose of tax codes that are also used for deductions. Another effective strategy has been the naming and shaming of tax delinquent individuals and firms (Angaretis et al., 2024; Dwenger and Treber, 2022), effectively increasing the general population's compliance rate through both reducing the wrongdoing of shamed taxpayer (*specific deterrence*) and for others through the prospect of getting shamed (*general deterrence*). However, this requires the tax agency to first inspect and identify which individuals and firms have conducted evasion, which might be limited given the low audit rates in the first place.

I estimate the effectiveness of an alternative policy instrument - *regulation based on visual signals* - on corporate tax evasion within the context of South Korea's license plate reform. South Korea introduced a new license plate regulation that mandates a distinct pale green color for corporate registered vehicles above a certain sales price. The new license plate policy aimed to deter individuals from registering luxury vehicles as corporate use and claiming undue tax benefits, which has emerged as a prevalent form of tax

evasion under South Korea's generous corporate tax code. The new regulation provides an ideal setting to investigate the effectiveness of an alternative policy tool when both third-party audits and name shaming are infeasible and the level of compliance monitoring remains low. This form of tax evasion is hard to uncover without in-person audits, while corporate tax audit rates continue to remain below 1% in South Korea (Republic of Korea, 2023).

I map the regulation using an action-signaling model (Butera et al., 2022), where the color of plates become an informative signal indicative of firm tax evasion. The new reform increases the visibility and informational content of license plates, reducing the image payoff for agents with a sufficient level of tax evasion prior to the policy. I mainly investigate responses to the license plate reform along the margin of new vehicle purchases, which I use as a proxy for unobserved tax evasion. Through a difference-in-difference event study design, I study whether the new license plate policy had a significant effect in reducing new corporate registered vehicle sales using non-corporate vehicles in the same price range as a control group. Moreover, I conduct a bunching analysis to analyze local responses around the price threshold to see any strategic sorting behavior to avoid new license plates.

Three key findings emerge from the empirical analysis. First, the new license plate policy had a significant deterrence event on new corporate vehicle sales as predicted by our theoretical model. The effect on new sales remain significant even after accounting for two months of anticipatory effects preceding the policy start date. The policy averted the sales of 9,401 new corporate vehicles above the price threshold during the first eight months of policy, which amounts to 2.2 months of pre-policy monthly average purchases in the same segment. Second, I also find evidence of local avoidance effects near the price threshold. Bunching estimates show that the excess mass for the price bin below the threshold is close to 100%, suggesting significant sorting of buyers to the left of policy threshold to avoid the license plate.

Lastly, the policy did not have a lasting effect on new vehicle sales with sales levels reverting to pre-policy levels after 8-10 months. This indicates that in the medium to long-run, the new license plate policy has failed to maintain its effectiveness at least in the external margin of new vehicle purchases. Based on our theoretical model, I hypothesize two possible explanations consistent with our observed phenomenon of short-term deterrence. Primarily, the proliferation of green plated vehicles may have reduced the relative visibility of each vehicle. Another possibility comes from the public recognition effect of distinctive license plates no longer continuing to become a factor of shame, even turning into a source of pride. Both explanations caution the need of carefully designing visibility and recognition parameters when designing signal-based regulations.

This paper contributes to three strands of literature. First, I contribute to the literature that empirically evaluates policies that tackle firm tax evasion. Traditional enforcement policies model tax evaders as rational actors, focusing on changes in audit rates and the amount of penalties to decrease the expected utility of evasion (Allingham and Sandmo, 1972; Becker, 1968). Recent research focuses more on the effectiveness of third-party verification in reducing the magnitude of tax evasion (Carrillo et al., 2017; Slemrod, 2019). I extend this literature by evaluating the short- and medium-run effectiveness of a novel policy, which can be utilized in tandem with, or even in the absence of above tools. Specifically, I provide empirical evidence on key parameters for implementation, as well as illustrate boundary conditions that affect long term deterrence.

Second, this paper adds empirical evidence on the effectiveness of signal-based policy instruments. Signal-based regulations can be defined as the government publicizing a coarsened indicator of regulatory activity to the public, which maps the unobserved behavior into easily assessable heuristics. Signal based regulations are widely used throughout different policy domains, such as restaurant hygiene grades (Jin and Leslie, 2003), school report cards (Andrabi et al., 2017; Bergman and Hill, 2018; Kolstad, 2013) and five star rating systems for Medicare and Medicaid (Vatter, 2025). I add to this literature by

adding another form of signal-based policy instrument that is more coarse but relies on directly attached visual cues to the regulated item of interest, which to my knowledge has not been studied.

Finally, my paper relates to a broader literature that documents the strategic response of consumers undermining policies that target vehicle purchase and registration. Previous research documents intertemporal substitution patterns before a policy comes into effect (Rittenhouse and Zaragoza-Watkins, 2018; Chen et al., 2026), which potentially confounds policy effects post implementation. My paper also documents a significant forward purchase of new vehicles along with local responses near the price threshold, suggesting that multiple margins that span both cross-sectional and time-series dimensions should be jointly analyzed in order to properly assess strategic behavior.

This paper proceeds as follows: Section 2 reviews the relevant literature. Section 3 discusses the theoretical model. Section 4 explains the policy and data. Section 5 presents the empirical strategy. Section 6 shows key results from our event study and bunching analysis. Section 7 discusses implications and Section 8 concludes.

2 Model of Public Recognition

Our setting is best explained by an action-signaling model (Butera et al., 2022). In this model, the agent cares how his action compares to the weighted average of population behavior when his action becomes publicly disclosed. The audience does not directly observe the agent's actions but instead receives signals related to true action, by which the audience utilizes to form a posterior expectation using Bayes' Rule.

Formally, define the level of action a $(a \in \mathbb{R}^+)$, types as θ $(\theta \in \mathbb{R}^+)$, signals as σ $(\sigma \in \{0, 1\})$ and updated posterior beliefs as $\mathbb{E}[a|\sigma]$. Firms choose the level of activity, which corresponds to the degree of illicit tax evasion through the use of corporate vehicles. Each firm's type can be defined as the underlying marginal propensity of engaging in such

activity. Similarly, $\mathbb{E}[a|\sigma]$ is the expected posterior belief of corporate tax evasion after observing the color of license plates. We define $G^*(\sigma)$ as the unconditional signal value distribution and H^* as the unconditional distribution of updated posteriors, $\mathbb{E}[a|\sigma]$. Note that both distributions $G^*(\cdot)$ and H^* are an equilibrium object, where $G^*(\cdot)$ is affected by equilibrium actions (a), distribution of types (F) and $G(\cdot|a)$ and H^* by G^*.

Define the payoff from public recognition:

$$\nu S\left(\mathbb{E}[a|\sigma] - \frac{\int_{a\in A} a\, w(a)\, dH^*(a)}{\int_{a\in A} w(a)\, dH^*(a)}\right) \tag{1}$$

where ν parametrizes the visibility of action, w is a smooth weighting function ($w : \mathbb{R} \to \mathbb{R}$) and S is a smooth and decreasing function with $S(0) = 0$.[1] Essentially, the public recognition payoff penalizes the discrepancy between expected level and a weighted benchmark (reference point) of actions, scaled by the function S and visibility of signals.

In equilibrium, each agent chooses a type-dependent action $\mathbf{a}(\theta) \in \mathcal{A}$ that maximizes

$$\Pi = \underbrace{\pi(a;\theta)}_{(1)\ Tax\ Benefits} + \underbrace{\phi}_{(2)\ OtherProfit} + \nu \int_\sigma \underbrace{S\left(\mathbb{E}[a|\sigma] - \frac{\int_{a\in A} a\, w(a)\, dH^*(a)}{\int_{a\in A} w(a)\, dH^*(a)}\right)}_{(3)\ Payoff\ from\ Public\ Recognition} dG(\sigma|a) \tag{2}$$

given $\mathbb{E}[a|\sigma]$ and its unconditional distribution H^*.[2] First two terms in equation (2) denote firm profits. Firm profits are decomposed into two categories: tax evasion through the use of corporate vehicles ($\pi(a;\theta)$) and overall firm profits irrespective of corporate car use (ϕ). More generally, $\pi(a;\theta)$ has a single peak in a and $\frac{d}{da}\pi(a;\theta)$ is increasing and bounded in θ. Thus each agent has an optimal intensity $a^*(\theta)$, with higher types enjoying higher payoffs from a higher level of a, which translates to more illegitimate firms having a

[1] The *pride* function in Butera et al. (2022) models $S(\cdot)$ as an increasing function. In this paper, I model $S(\cdot)$ as a decreasing function, or equivalently, a *shame function* as the underlying action is undesirable.

[2] The action-signaling model of regulation bears relevance to canonical models of deterrence (Allingham and Sandmo, 1972; Becker, 1968), where crime (or evasion) is understood as a rational decision made after comparing the expected benefits and cost. In those models, the level of evasion is parameterized as a function of (i) probability of detection and (ii) expected penalties. In our model, increasing visibility (ν) closely relates to the probability of detection, while the agent chooses the level of evasion that maximizes expected payoffs as in the Becker model.

higher marginal benefit from tax evasion.

Uninformative Signals Prior to the new plate policy, existing license plates did not allow for any inference or updating on the degree of expected corporate tax evasion, as corporate vehicles were not visually distinguishable. In this case, $\mathbb{E}[a|\sigma]$ becomes degenerate to $\bar{a}$[3] and $H^*(\cdot)$ puts a unit mass on $\bar{a}$. The public recognition term on equation (2) cancels out as $\bar{a} - \bar{a} = 0$ and $S(0) = 0$. Under our assumption of single-peaking of $\pi(a;\theta)$, reducing tax evasion decreases payoffs under the previous scheme. Thus evading firms do not have any incentive to alter their current action.

$$\Pi_i = \underbrace{\pi(a;\theta)}_{(1)\ Tax\ Benefits} + \underbrace{\phi_i}_{(2)\ OtherProfit} + \nu \int_\sigma \underbrace{S\left(\mathbb{E}[a|\sigma] - \frac{\int_{a\in A} a\, w(a)\, dH^*(a)}{\int_{a\in A} w(a)\, dH^*(a)}\right)}_{(3)\ Payoff\ from\ Public\ Recognition} dG(\sigma|a) \quad (3)$$

Model Prediction New policy increases visibility of corporate-registered vehicles through visual markers and sends a more informative signal of evasive activity due to the price threshold.[4] New license plates, *ceteris paribus*, will largely increase visibility as the pale green color was intended to make corporate vehicles easily distinguishable. Furthermore, the price threshold applied for green plates makes clear that the vehicle falls into the luxury vehicle segment. Assuming that tax evasion occurs with a higher frequency and intensity in upper class models, receivers will shift their posterior upward compared to the general expected tax evasion of the entire corporate population, $\bar{a}$. For a firm with sufficiently high level of tax evasion, this leads to a reduction evasive activities as the public recognition term in equation (3) turns negative.

Since the new regulation decreases expected payoffs, we would expect avoidance responses on the external and internal margin of vehicle purchase, as well as the usage

[3]On the other hand, $\mathbb{E}[a|\sigma] = a$ if signals are fully informative.

[4]Note that signals are not directly tied to the agent itself (i.e. the observer observes the color of plates but does not observe which firm is owning the vehicle). While directly tying signal to the agent is not necessary for the policy to achieve the intended effect, it will have implications for the behavioral mechanism. Specifically, linking signals with agents would have a stronger behavioral effect on the tied agent (specific deterrence) compared to signals that are not tied.

behavior for existing vehicle owners. On the external margin of purchase, corporate-registered purchase would decline for vehicles above the price threshold. Conditional on purchasing, we would expect bunching for corporate vehicles left to the threshold as a discontinuous drop in expected profits occur. Another possibility is purchasing a similar model but not registering as a corporate vehicle. This, however, is less likely to occur given the magnitude of tax benefits that follow from corporate registration. Lastly, owners of corporate vehicles could drive less or shift the timing of driving to reduce exposure, and prolong the existing fleet of corporate cars. We empirically investigate the first and second possibility in our analysis.

3 Data

3.1 Policy Background

South Korea's corporate tax code provides generous tax benefits to vehicles registered as corporate use to facilitate corporate economic activity. Primarily, the current tax code allows a maximum of 8 million KRW (approximately $5,926)[5] per year for each corporate registered vehicle as to be counted as depreciation expense for the first five years (Republic of Korea, 2023). In addition, other maintenance costs such as fuel cost, insurance fee and vehicle registration fee can be accounted as corporate expenses until 15 million KRW (approximately $11,538),[6] without any further proof.[7] Relevant costs that exceed the 15 million cap can also be accounted as business expenses without limit as long as firms provide a proof of of business-related vehicle use. Since taxable profits are determined from firm revenue net of expenses, this effectively reduces corporate tax for a given tax rate, or bring the level of taxable profits down to a lower tax bracket.

[5] I use an exchange rate of 1 USD = 1,350 KRW throughout the paper.

[6] This includes depreciation expenses.

[7] The only requirement to claim above corporate tax benefits is enrolling in employee-only auto insurance. This, however, is considered to be less of a binding constraint as all drivers should opt in to an insurance policy regardless of corporate registration.

Corporate registered vehicles are mainly utilized for daily business purposes in South Korea such as for business trips and for visiting clients.[8] Also, major conglomerates configure chauffeur driven vehicles for high rank executives, where the vehicle segment and displacement generally increases with rank. For example, Samsung Electronics reserves a 3.0-liter engine vehicle (Hyundai Grandeur) for Managing Directors, a 3.5-liter engine vehicle (Hyundai Genesis or Kia K9) for Senior Managing Directors, vehicles with a 4.0 liter-engine or below for Vice Presidents, and 5.0-liter engine vehicle for Presidents, where executives including and above the rank of Vice President having an option to choose foreign imported vehicles such as a Mercedes or a BMW (Kang, 2021). Note that major firms primarily configure Korean vehicles rather than foreign imported cars except for the highest ranked executives.

However, there were increasing investigations from Korea's National Tax Service where members of corporate ownership have purchased luxury or sports cars as corporate-registered vehicles but instead utilized them for personal use, dubbed by the agency as "Young & Rich Tax Evaders" (National Tax Service, 2021). Individuals utilizing corporate registered vehicles for non-business purposes directly violate the Korean tax code in two ways. Primarily, this would constitute corporate tax evasion, as these vehicles are used to falsely claim corporate tax benefits without any proper business use. Second, if corporate vehicles are mainly utilized as personal use for corporate members, this is generally viewed as a form of bonus payment that should be subject to income tax and other withholding tax were it to be properly documented.

Starting from January 1st, 2024, Korean government mandated all corporate-registered cars above 80 million KRW (approximately $59,260) to carry a distinct license plate "to avoid private use of corporate-registered vehicles and avoid tax evasion" (Ministry of Land, Infrastructure and Transport, 2023). Figure 1 shows a visual example of white li-

[8]Green-plated vehicles do not generally display the identity of firms, so the firms owning corporate vehicles are generally not inferrable by bystanders. However, they will be noticed by business counterparts or partnering firms if used during business purposes.

cense plates and new green plates. The license plate featured a pale, light green color designed to increase public visibility and visually distinguish regulated vehicles from other vehicles, which typically carries a white license plate or a pale blue license plate (only for electric vehicles).[9] The price threshold is typically used by vehicle insurance providers to differentiate the luxury vehicle segment and charge additional premiums (Ministry of Land, Infrastructure and Transport, 2023). The regulation contains aspects of voluntary regulation, as it does not consider prior records of tax evasion or directly aim for stigmatization but generically applies to all vehicles if the price and corporate criterion are met, "to encourage firms to appropriately manage and use their vehicles with designated business purposes" (Ministry of Land, Infrastructure and Transport, 2023). The new regulation applies to both new and used vehicle market and leased or rented vehicles but does not apply to existing vehicles.

Table 1: Target of Regulation

	Price ($\geq$ 80m)	Price ($<$ 80m)
Non-corporate	no change	no change
Corporate	**Regulated**	no change

Note: This table demonstrates the target of South Korea's new license plate regulation based on two dimensions: (i) Price of purchase (in KRW) and (ii) whether a vehicle was registered by a corporate entity or not.

Several features of the new policy should be noted. First, the policy threshold applies to the final price reported by car dealers rather than a pre-determined price indicator such as Manufacturer Suggested Retail Price (MSRP). Dealers have substantial discretion in determining the final sales price by charging different markups, selectively applying incentives and rebates and bargaining between customers. Thus final transaction prices in

[9]Prior to this regulation, license plates of corporate and non-corporate vehicles were largely indistinguishable. Existing license plates comprised of eight letters, which is a combination of three number digits + one Korean character + four number digits in this respective order (refer to Figure 1 for an example). Rented or short-term vehicles had to carry one of three specific Korean characters in their license plate. However, long-term lease vehicles or vehicles directly purchased by firms were plated similarly to the rest of the vehicle fleet.

Figure 1: Comparison of Existing and New License Plates

Note: Above figure shows white license plates (left) used by vehicles prior to the regulation and green license plates (right) used by corporate-registered vehicles above the price threshold post-policy.

principle may vary along customer and season even for a given dealer-make-model pair. Combined with the voluntary reporting of prices from dealers, this opens the avenue for dealers and corporate customers to manipulate the final transaction price to avoid the new policy, especially for models priced near the threshold. Second, the new policy was announced approximately two months prior to implementation, on November 3rd, 2023. While this makes policy predictable and buys time for administration, this also allows strategic buyers to exploit this temporal window and shift vehicle purchase forward before the new policy comes into effect.

3.2 Data and Descriptive Statistics

To identify the new license plate policy's effect on sales of corporate registered vehicles, I construct a unique model-level dataset that combines two sets of official data. Table 2 shows the summary statistics based on vehicle purchase price.

New Vehicle Registration Data I obtain the universe of all foreign-imported vehicles registered one year pre- and post- policy, obtained through an information request at South Korea's Ministry of Land, Infrastructure and Transport. This dataset consists of transaction level data from January 2023 to December 2024 on (i) month and year of registration (ii) make and model (iii) whether the vehicle was registered as corporate or not

Table 2: Summary statistics

	Mean	Median	SD	Min	Max	N
All Vehicles						
Price	79.902	63.182	50.756	3.382	1,092.727	551,810
Registration						
Corporate	99.235	75.774	64.401	5.000	1,092.727	204,263
Non-corporate	68.539	59.900	36.108	3.382	1,048.600	347,547
Vehicle Origin						
UK	100.750	44.091	107.472	5.000	1,092.727	28,687
France	35.733	36.433	4.653	19.102	64.658	3,125
Italy	331.045	355.669	166.190	20.636	1,085.418	2,311
US	58.497	52.515	19.595	18.182	167.711	73,957
Germany	85.971	68.636	45.158	3.382	941.285	359,574
Japan	58.735	59.455	17.353	26.883	368.265	49,634

Note: Above figure displays summary statistics on the final price of sale. All columns except the number of observations (N) is measured in million KRW. First panel summarizes the price variable for the entire sample. Second panel shows statistics for the same variable based on corporate registration status. Third panel divides the sample based on the origin of vehicles.

(iv) city and district level of registration and (v) final purchase price of transaction. Since I observe all vehicles imported during our period of interest, I do not expect to suffer from missing data or data selection issues. Also, note that observing the final transaction price yields more internally valid estimates compared to other metrics of price, as the government utilizes final sales prices to determine plate color. A limitation of our data is that while we observe a rich set of data on the vehicle that was being sold, we were not able to extract information on the specific characteristics of firms that conducted the purchases due to privacy concerns.

Descriptive Patterns The prevalence of luxury vehicles for corporate use per se may not indicate any wrongdoing. Nevertheless, raw patterns from our data suggest that certain motives beyond ordinary business activity exist in these purchases.

Figure 2 plots an histogram of total corporate- and non-corporate vehicle sales by 10 million KRW (approx. $7,407) price bins during our period of interest. Gray line displays the mean corporate registration ratio for each price bin, defined as corporate regis-

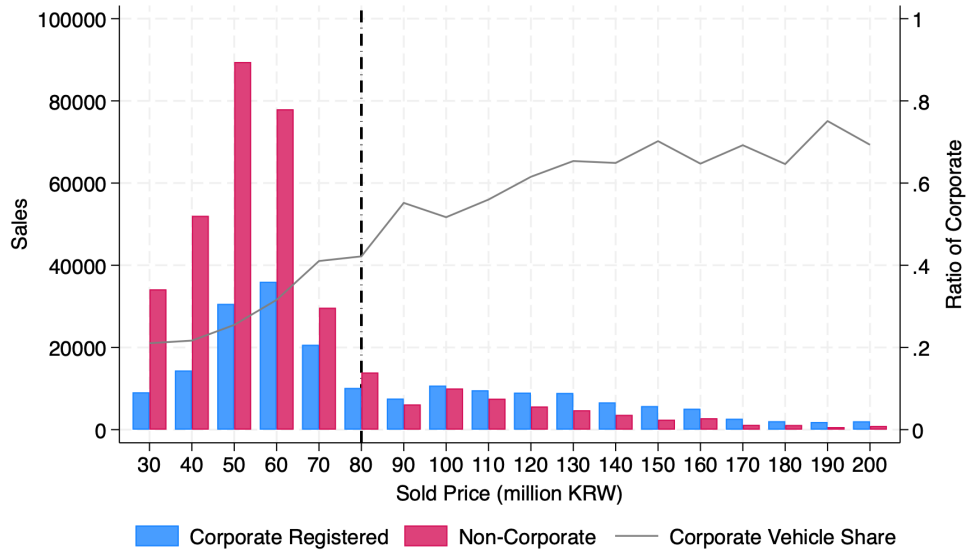

Figure 2: New Sales Count and Corporate Registration per Price Bin

Note: Above figure shows the sales count of corporate and non-corporate registered vehicles during our total sample period (2023-2024) for each 10 million KRW price bin. Blue color denotes corporate-registered sales and red color denotes non-corporate sales, respectively. Grey line connects each price bin's corporate share rate, computed as the ratio between corporate sales count and total sales count for each price bin. Dashed line indicates the policy price threshold at 80 million KRW. New vehicles with sales price below 30 million KRW are recorded as 30 million.

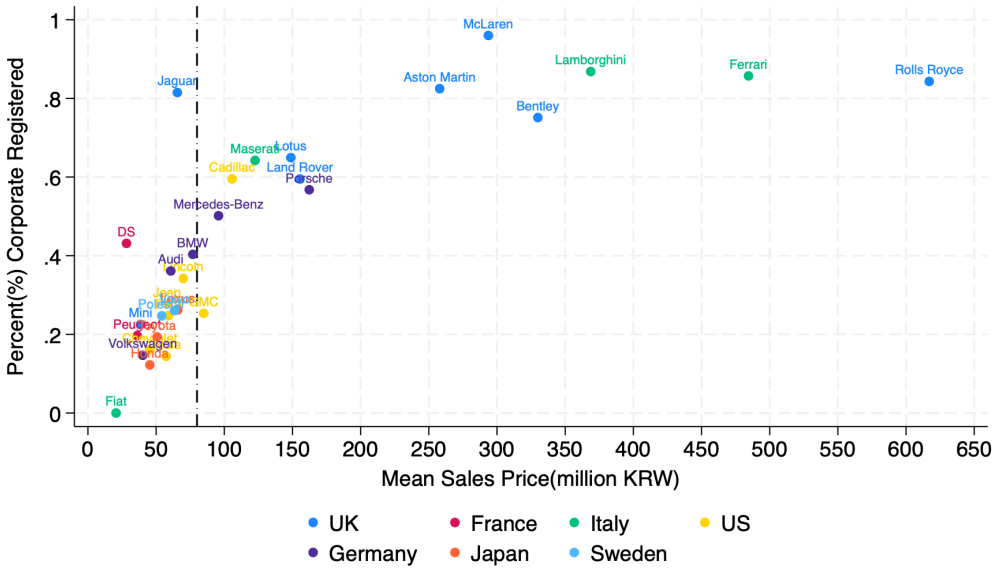

Figure 3: Average Sales Price and Corporate Vehicle Ratio and per Automaker

Note: Above figure plots the average sales price (horizontal) and proportion of corporate registered vehicles (vertical) by each automaker observed during our total sample period (2023). Each dot's color indicates automaker's country of origin. Dashed line indicates policy price threshold (80m KRW)

tered sales over total sales. Non-corporate sales dominate in the lower segments of the new vehicle market. For price bins below our policy threshold, corporate registration ratio remains lower than 50%. However, this trend reverses as we shift to the more expensive vehicle segments. The share of corporate purchases continue to increase with price, with corporate purchases nearing 80% in vehicles including 200 million KRW (approx. $148,148) or above. Assuming diminishing marginal returns associated with vehicle price, observed patterns are largely inconsistent with cost minimizing behavior of firms. Appendix Figures A1 and A2 demonstrate similar patterns for pre- and post- policy periods separately.

Figure 3 plots the average corporate registration rate of each manufacturer's new sales during our sample period, along with mean sales price for each automaker plotted in the horizontal axis. A strikingly similar pattern emerges with respect to the previous figure. As we move beyond the policy threshold and into the more expensive price regions, the percentage of new vehicles registered as corporate use increases, with corporate registration rate exhibiting a near logarithmic relationship with mean prices. The relationship between mean price and corporate registration rate does not seem to be affected differently by the type of model produced by manufacturers. 85.7% of Ferrari cars in our sample are registered as corporate vehicles, whereas 86.8% of Lamborghini cars are corporate registered. This further lends support to the existence of non-business motives, as some vehicles (ex. sports cars) are less likely to be utilized in day-to-day business activities. Again, Appendix Figures A3 and A4 demonstrate similar patterns for pre- and post-policy periods, ruling out the explanation that pre-policy patterns drive the result.

4 Empirical Strategy

Our unique research setting generates several inter-temporal and cross-sectional substitution margins along potential buyers that pertain to identification. First, potential

buyers in the regulated segment could shift purchases forward before the policy comes into effect, taking advantage of the announcement period. Second, potential buyers could continue to purchase corporate vehicles post-regulation but shift to a lower price segment in order to avoid green plates. Third, a similar local substitution pattern could occur post-policy to the purchase of used vehicles below the price threshold. Fourth, buyers in the regulated segment could still remain in the regulated segment but purchase used vehicles instead of new vehicles. Finally, buyers could forgo corporate tax benefits entirely and instead purchase non-corporate vehicles in a similar price segment, avoiding green plates.

This paper primarily focuses on the new vehicle market and aims to quantify the effect on new vehicle purchases taking anticipation into account (first margin) and investigate local responses in the regulated segment near the price threshold (second margin). Considering all possible margins of substitution would be important to comprehensively assess the new license plate policy, given the importance of used car markets as a close substitute to the new car market (Gruenspecht, 1982) and that we proxy vehicle purchases for unobserved tax evasion. However, data constraints prevent the analysis of the used car market. Thus we mainly analyze the new vehicle market, which would more consistently reflect demand and would be the largest margin affected by the policy considering the lifecycle of vehicles (Gillingham et al., 2022).

4.1 Difference-in-Difference Event Study

To estimate the effect of the new license plate policy on corporate-registered luxury vehicle sales, I utilize a difference-in-difference event study model using the sales of non-corporate vehicles above the price threshold as a control group.[10] Generally, I would to

[10] I estimate an event-study instead of a general difference-in-difference(DD) estimator, because an event study model allows for monthly estimates while DD estimators average out temporal variation within a given pre- or post- period. This distinction bears importance in our setting given the durable goods nature of vehicles and also the presence of anticipation. Since newly added cars remain on the vehicle fleet for a significant amount of time, the cumulative stock for each pre- and post- period becomes relevant, rather than

estimate the following difference-in-difference event study estimator:

$$y_{ct} = \alpha_c + \tau_t + \sum_{j \neq -1} \gamma_j \cdot \mathbf{1}\{c = 1\} \cdot \mathbf{1}\{T - t = j\} + \varepsilon_{ct} \tag{4}$$

where y_{ct} denotes aggregate monthly sales per corporate (and non-corporate) vehicles and month, α_c denotes a dummy for corporate registration, τ_t month fixed effects, $\mathbf{1}\{c = 1\}$ an indicator for corporate registration and T denoting the first period of interest. Several issues arise with this approach. A key restriction is a low degree of freedom which renders the monthly event study model collinear due to our high degree of aggregation. Another concern comes from the inability to account for model-level shocks that affect purchasing decisions at the buyer level .

Instead, I execute an event study at the make-model level, where I calculate the policy parameters of interest based on each vehicle model's total sales and back out aggregate monthly sales. Appendix B shows how our model-level approach closely recovers point estimates compared to the aggregated model.

My preferred functional form is the following:

$$y_{mct} = \alpha_c + \tau_t + \sum_{j \neq -1} \gamma_j \cdot \mathbf{1}\{c = 1\} \cdot \mathbf{1}\{T - t = j\} + \varepsilon_{mct} \tag{5}$$

where all other notation follows equation (4), but y_{mct} now denotes the monthly count for each model's corporate-registered vehicle sales. Note that to approximate aggregate sales, α_c represents a corporate registration dummy rather than model fixed effects (α_m). Also, we define treatment timing (T) as the month of policy announcement as opposed to the month of implementation to account for anticipation effects. Our coefficient of interest is $\hat{\gamma}_j$. Under the assumption of parallel trends and no anticipation in $T < -1$, $\hat{\gamma}_j$ identifies the per-month causal effect of the new policy on new corporate vehicle sales over the

average per-period purchases. Moreover, DD estimators lead to upward (downward) bias in the presence of anticipatory purchasing when we set the treatment start date as the announcement (implementation) date.

threshold. We expect per-month coefficients $\hat{\gamma}_j$ to be statistically indistinguishable from zero for pre-announcement periods ($j < -1$).

Control Group Validity The use of non-corporate purchases as a control group further warrants discussion. Firms and individuals are in practice, subject to distinct preferences, different financial constraints or seasonal demand patterns, directly challenging the assumption that both groups would have exhibited a similar behavioral pattern in the absence of policy.

I discuss the above concern in two ways. First, I should note that this paper only focuses on the foreign-imported vehicle market as opposed to the entire vehicle market in South Korea, which nets out a substantive portion of genuine business purchases that may differ from individual purchases. While corporate vehicles in the domestic market are generally viewed as legitimate corporate use (Kang, 2021), the motive behind foreign-imported luxury vehicles purchases are viewed to be less tied to true business motives, which in turn motivated the new policy itself. This indicates that a meaningful proportion of corporate vehicle demand in the luxury segment, considering their non-business purposes, is analogous to the non-corporate buyers. Second, Chen et al. (2026) analyzes anticipatory purchasing in Sweden's feebate policy and demonstrates that the vehicle purchasing behavior of firms and non-firms exhibit a similar pattern. Utilizing residuals from predicted sales, both individuals and non-retailer firms move in a similar direction and magnitude, further suggesting that firm and non-firm differences pose less of a threat to validity when analyzing new vehicle purchase.

4.2 Bunching Analysis

To gauge avoidance behavior of new corporate vehicle buyers near the policy threshold, I analyze whether the mass of new sales bunch below the price threshold after the

policy comes into effect. Bunching estimators generally take the following form:

$$\eta_i = \sum_{k=0}^{K} \beta_k p_i^k + \sum_{r \in [-R,R]} \rho_r \mathbf{1}\{p_i = r\} + \varepsilon_i \tag{6}$$

where η_i denotes the number of observations in a predetermined bin i, p_i^k each bin's mid-point value, $[-R, R]$ narrow bandwidth near the bunching threshold and $\mathbf{1}\{p_i = r\}$ an indicator function of whether each bin falls within the bandwidth. Similar to the notion of donut regressions, above specification approximates the observed bin frequency with a flexible polynomial function, excluding (or equivalently, completely absorbing) a narrow region near the threshold through the use of fixed effects. Under the assumption that (i) manipulation is one-sided and bounded and (ii) the counterfactual distribution follows general regularity conditions, the counterfactual density of the excluded region can be recovered through fitted values of equation (6).

$$\hat{\eta}_i = \sum_{k=0}^{K} \hat{\beta}_k p_i^k \tag{7}$$

In practice, I approximate the density function from 70 million KRW to 90 million KRW using a third-degree polynomial, excluding the region of $\pm$ 2 million KRW surrounding the policy threshold of 80 million KRW.

$$\eta_i = \sum_{k=0}^{3} \beta_k p_i^k + \sum_{r \in [-78,82]} \rho_r \mathbf{1}\{p_i = r\} + \varepsilon_i \tag{8}$$

Using fitted values from the polynomial in equation (8), I construct the counterfactual mass for the excluded regions $r \in [-78, 82]$. I then calculate the excess mass for each price bin as the difference between observed an counterfactual mass, noramlized by the counterfactual mass:

$$b = \frac{(Observed\ Mass) - (Counterfactual\ Mass)}{(Counterfactual\ Mass)} \tag{9}$$

5 Results

5.1 Difference-in-Difference Event Study

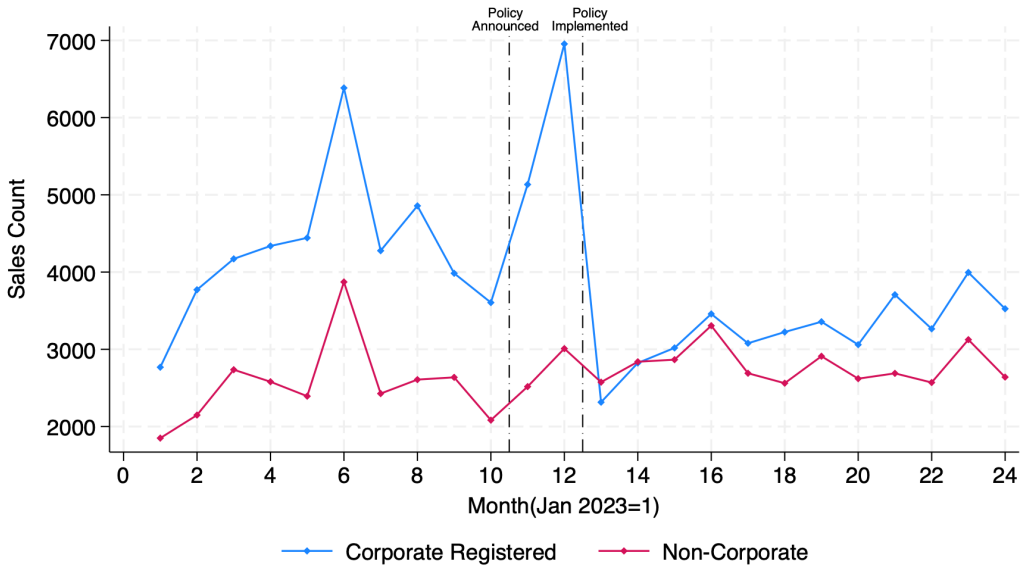

Figure 4: Aggregate Monthly Sales for Corporate and Non-corporate Vehicles

Note: This figure plots the total monthly sales count for corporate-registered vehicles (blue) and non-corporate vehicles (red). Horizontal axis displays month indicator with January 2023 normalized as the first month. Dashed lines indicate the month when policy was first announced (Nov 2023) and implemented (Jan 2024), respectively.

Figure 4 plots aggregate sales patterns for the treatment and control group during our period of interest, which covers a total of two years with one year pre- and one year post-policy. I mark new monthly sales for corporate-registered vehicles above the price threshold in blue and non-corporate vehicles in red. Also, I indicate the period when policy was announced (November 2023) and implemented (January 2024) in dashed lines, respectively. Prior to policy announcement, both treatment and control groups follow a similar trajectory, with the treatment group recording approximately 2,000 more sales compared to the control group. This relationship changes after the policy becomes announced but yet to be implemented, as corporate vehicle sales above the price threshold exhibit a sharp increase. This spike in corporate sales is followed by a steep sales decline

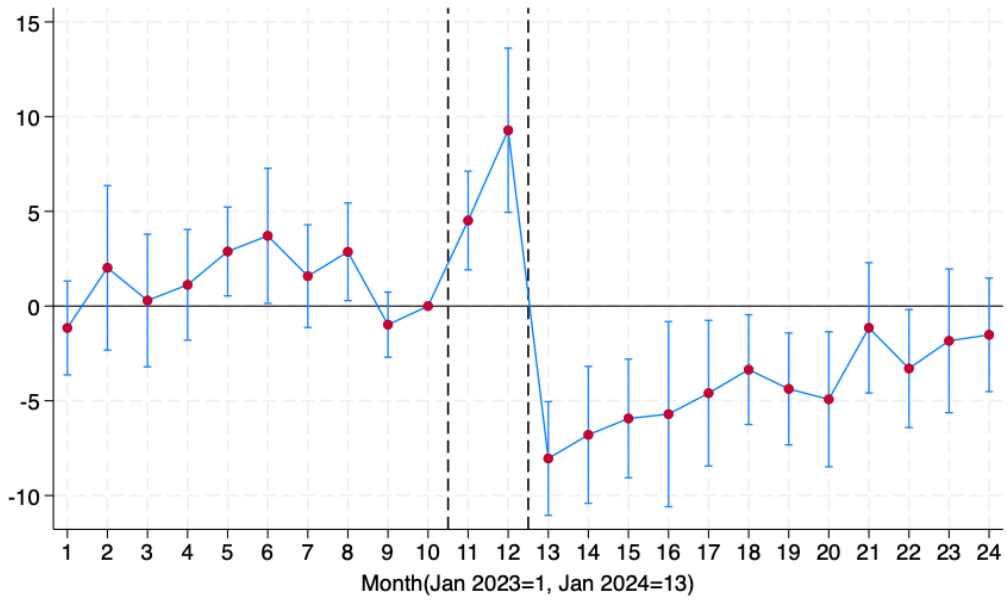

Figure 5: Event Study Estimates

Note: Above figure plots point estimates and 95% confidence intervals for each month from equation (5), with one month prior to policy announcement (Oct 2023) as the omitted reference period. Horizontal axis displays month indicator with January 2023 normalized as the first month. Dashed lines indicate the month when policy was first announced (Nov 2023) and implemented (Jan 2024), respectively.

as the new license plate policy gets implemented.

Figure 5 displays our main event study results from equation (5), which allows for a detailed month-by-month comparison between corporate and non-corporate sales. Our preferred threshold for sample selection is 75%: that is, we include models where 75% of sales occurred with a price including and higher than the policy threshold (80 million KRW) during our pre-policy period.[11] Comparing the aggregate sales pattern (Figure 4) and patterns from the event study (Figure 5) suggests that our model-based approach closely recovers the aggregate sales count. While several months during our pre-policy periods exhibit a statistically significant difference between the treatment and control group, I cannot reject a joint F-test that assumes linear trends are statistically different from zero (Pr>F 0.98). This provides suggesting evidence for the parallel trends assumption which underpins our casual identification.

[11] Appendix Figure A7 shows that using other thresholds - 90% or 50% - yields quantitatively similar results

Point estimates suggest that publicizing the policy in advance led to 1,418 additional corporate purchases above the price threshold in November and 2,914 purchases in December,[12] which corresponds to 33.3% and 68.5% of pre-policy monthly average purchases in the regulated segment, respectively. However, corporate sales sharply declined after the policy went into effect. Corporate purchase for luxury cars decreased 2,526 on the first month post-policy (January 2024) and 2,134 on the second month (February 2024), which correspond to 59.3% and 50.1% of pre-policy monthly purchases. Importantly, the new license plate policy only had a short-term effect on new vehicle sales, with corporate sales reverting to pre-policy levels 8 to 10 months after the policy went into effect.[13] Appendix Figure A8 and A9 shows that the effects are primarily driven by German-imported vehicles, which account for about 65.2% of our sample. Post-policy coefficients for non-german imported vehicles are statistically indistinguishable from zero.

Accounting for Anticipation To accurately assess policy effects in our setting, we need to assess whether observed decline in sales post-implementation is higher in magnitude net of the observe sales increase during the anticipation period.[14] Since vehicles are a durable good that lasts at least several years, sales spike prior to the purchase affects the probability of purchase during the policy. If the magnitude of both forces are comparable, this would indicate that only the timing of purchase has shifted forward rather than a true decline in sales, close to a time-based notch. In this case, the true amount of new sales for the regulated class for a given period would be close to zero, suggesting a null effect of policy on new sales.

[12] Appendix B shows that $\hat{\beta}_{month} \times N$ approximates point estimates for each month into total sales.

[13] The short-lived nature of policy effect echoes the findings from the Cash-for-Clunkers program, which provided consumer rebates for purchasing a new fuel efficient vehicle to pull purchases forward. See Hoekstra et al. (2017); Mian and Sufi (2012); Li et al. (2013)

[14] There are three approaches that detect and distinguish unobserved anticipation from other confounding factors, such as endogenous treatment. First, Malani and Reif (2015) fully parameterizes anticipatory behavior into a forward-looking utility model. Second, another line of research utilizes observable predictors of behavior and analyzes residuals assuming conditional unconfoundedness to detect anticipation(Chen et al., 2026; Lam and Bausell, 2007; Rittenhouse and Zaragoza-Watkins, 2018) Third, Alpert (2016) utilizes a difference-in-difference approach to compare drugs that are more likely to exhibit anticipatory purchase compared to less likely drugs.

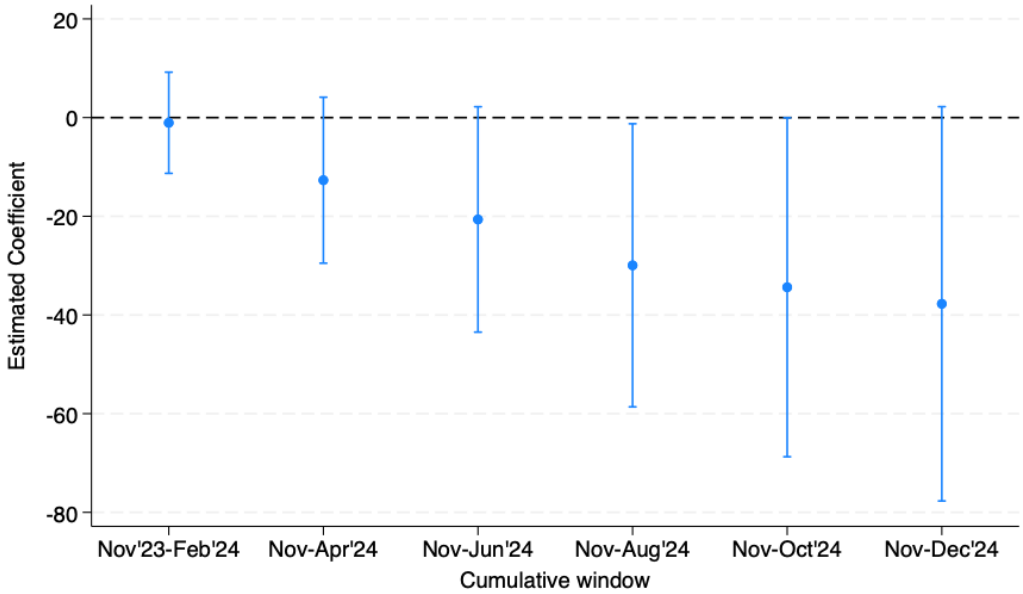

Figure 6: Policy Effect Accounting for Anticipation

Note: Above figure plots the linear combination point estimates and 95% confidence intervals, derived from monthly event-study coefficients based on equation (5) and varying the coverage of coefficients of interest. The first result plots the linear combination of monthly coefficients from November 2023 to February 2024. Remaining results combine monthly coefficients as noted.

To empirically investigate this matter, I compare the sum of monthly event-study coefficients during the anticipation period and post-implementation period, varying the length of months included in the latter period.[15] Figure 6 displays the results accounting for anticipatory effects. I include a linear combination of the two-month anticipatory period and two months, four months, six months, eight months, ten months and twelve months post-policy, respectively. For example, the first column shows the point estimates and confidence intervals of $\hat{\gamma}_{Nov} + \hat{\gamma}_{Dec} + \hat{\gamma}_{Jan} + \hat{\gamma}_{Feb}$ from equation (5). My preferred speci-

[15]Fully attributing monthly effects as anticipation assumes that absent announcement, both treatment and control groups would have moved in a similar direction. Following the notation of Rambachan and Roth (2023), we can decompose the true monthly coefficient vector β into treatment effect (τ) and differential trends between treatment and control (δ) in the absence of policy. $\beta = \begin{pmatrix} \tau_{\text{pre}} \\ \tau_{\text{announce}} \\ \tau_{\text{post}} \end{pmatrix} + \begin{pmatrix} \delta_{\text{pre}} \\ \delta_{\text{announce}} \\ \delta_{\text{post}} \end{pmatrix}$ No treatment effect pre-announcement imposes $\tau_{pre} = 0$ and parallel trends assumption imposes $\delta_{post} = 0$. I additionally impose that $\delta_{announce} = 0$ holds for the two months when the policy was announced but not implemented, which allows for the identification of policy effect during the announcement period, $\tau_{announce}$.

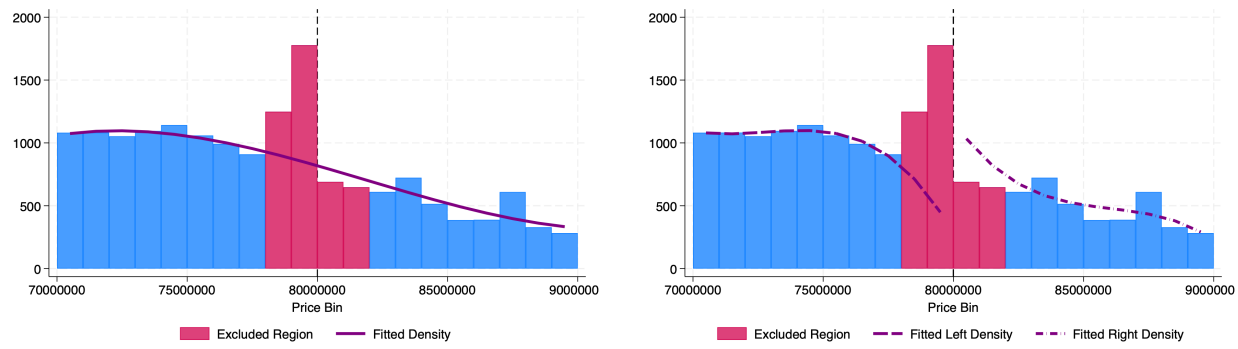

Figure 7: Bunching Histogram and Fitted Lines

Note: Each figure plots the histogram of new corporate-registered vehicle sales after the policy was in effect and a fitted line, for vehicles with sales price between 70 million KRW and 90 million KRW. Histograms in red denote the excluded region during our bunching estimation. Fitted lines use third-degree polynomials from equation (8). The left figure fits a polynomial using all data between 70 million and 90 million price bins excluding 78-82 million KRW region. The right figure fits polynomials separately left and right to the price threshold. Dashed lines in black indicate the 80 million KRW policy price threshold.

fication is the eight-month post-implementation window covering up to August 2024, the last month when the policy had a statistically significant effect during our event study.

The new license plate policy brought a statistically significant decline in luxury corporate vehicle purchases even after accounting for anticipatory effects. Point estimates suggests that the policy brought a cumulative decline of 9,401 corporate-registered vehicles above the price threshold compared to non-corporate vehicles, or 2.2 months equivalent of average monthly pre-policy sales in the same segment. Point estimates remain relatively stable for later columns that include a wider timeframe such as November 2023 until October 2024 (fourth column) and November 2023 until December 2024 (fifth column), consistent with null effects in our event study from month 9-10 onward. The large confidence intervals though, should be noted and cautioned when drawing further conclusions.

5.2 Bunching Analysis

Figure 7 plots the histogram of new sales around the price threshold post-policy. Histograms in red denote the excluded region during our estimation and lines denote the

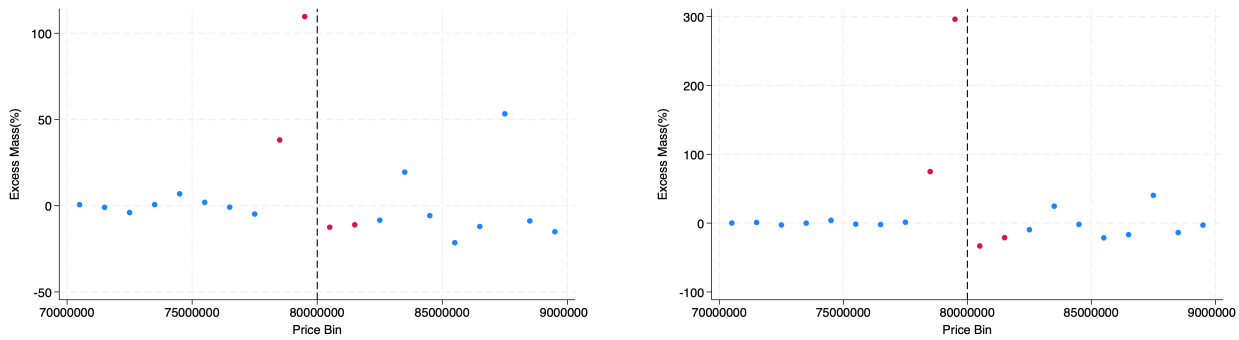

Figure 8: Excess Mass Calculation

Note: Each figure plots the excess mass calculated using equation (9). The left figure plots the excess mass using polynomials fitted through the entire 70m-90m region and the right figure plots the excess mass fitted separately to the left and right of the threshold. Dashed line in black indicates the 80m KRW policy threshold. Red dots indicate four price bins near the price threshold that are excluded during the bunching estimation.

fitted density counterfactual density using our preferred specification. Left figure plots a global third-order polynomial and the right figure plots third-order polynomials separately on each side of the price threshold. The histogram clearly shows sorting to the left of the price threshold. Price bins directly below the threshold exhibits a distinctively larger mass compared to other price bins, which show a smooth decline as the final sales price increases.

To quantify the extent of bunching, Figure 8 plots the excess mass of the histogram relative to our counterfactual density. Our preferred specification is the approximation using global polynomials, shown in the left graph of Figure 8. Results show that compared to an expected, smooth density, price bins just below the threshold exhibit more than 100% extra mass. This indicates that to avoid carrying a green license plate, buyers strategically adjust the final sales price or opt to purchase a vehicle with a final price less than, but not too far from, the threshold of 80 million KRW. The righthand side of the threshold, however, does not exhibit a corresponding missing mass proportional to the bunching behavior and has a density close to its expected mass. This indicates that the excess mass just left to the threshold comes from not just the price bins above the threshold but from a broader region throughout the right side.

Similar pattern emerges with the graph on the right side of Figure 8, using a separate polynomial each side to the threshold. This graph has a higher prediction accuracy where most of the expected mass does not deviate largely from the observed mass. The implied excess mass is around 300% for the price bin directly left of the threshold and close to 100% for the next price bin to the left. However, the implied magnitude requires caution as fitted graphs in Figure 7 are less smooth and tend to overfit near the threshold, compared to the global polynomial.[16]

6 Discussion

Our empirical results show that the new license plate policy deterred firms from purchasing luxury corporate vehicles and firms strategically purchase vehicles below the threshold to avoid the new plate. Both results are consistent with model predictions, which hypothesizes the loss in firm payoffs from the new license plate policy. However, policy effects last only for 8-10 months and vanish afterwards. This suggests that in the medium to long run, the policy fails to maintain the initial deterrence effect, requiring a more nuanced evaluation of its effectiveness. One caveat is that our analysis only tackles two observable phenomena among a range of predicted actions that also bears relevance when assessing policy effects. Thus our analysis on new vehicle sales would be a sufficient but not a necessary condition for determining policy effectiveness. In this section, I lay out two alternative explanations consistent with a short-term effect but no long-term effects.

First, an increasing number of green plated vehicles decreases the *relative* visibility for a single vehicle, thereby attenuating the shaming penalty term in equation (2). In the first few months of the policy, the share of green plated vehicles among the vehicle fleet remains low. However, this number gradually increases as more and more vehicles

[16] Appendix Figures A5 and A6 provides suggestive evidence that although with a smaller magnitude, bunching behavior continues throughout the post-policy period in our sample.

above the threshold are sold each month. From a single firm's perspective, the relative probability of getting noticed becomes lower as all vehicles over the threshold carry the same plate design with the same green color. Note that I am interpreting the visibility parameter (ν) as a relative rather than an absolute term.[17] This would be a consistent explanation with our action-signaling model.

A second explanation comes from the misspecification of our shame function, $S(\cdot)$. Anecdotal evidence suggests that the green license plate no longer acts as a red flag, but rather a status symbol that indicates family wealth and firm ownership. This phenomenon would indicate that the shame function is no longer decreasing for *all* agents or firms. Rather, this suggests that for some firms, the public recognition function $S(\cdot)$ is increasing or equivalently, works as a *pride function*. In this case, monthly event study parameters would reflect a composite effect and depend on whether there are more potential buyers that perceive the license plate of a shame than pride. Either way, this suggests that our action-signaling model is misspecified.

Both explanations suggest the necessary conditions for a signal-based regulation to be effective. First explanation suggests the need to control how visible a signal would be, accounting for relative visibility as well. This suggests that while the color scheme may be a good policy instrument, confining the number of vehicles that carry the distinct color to a small number that does not increase over time would increase the policy effect. For example, this could be done by increasing the price threshold to a higher price region, or only mandating plates for firms that have a prior record or higher probability of wrongdoing.

On the other hand, second explanation suggests that signal-based regulations should provide a clear and succinct direction on how the regulation should be interpreted. Opening more than a single avenue for the public to interpret signals would not bring a long-lasting effect, or even a short term effect after all. One good example of a distinct signal

[17] While the original action-signaling model does not make any distinction of whether this parameter should also be translated in a relative way, the comparative and relative nature of shame and pride lends credibility to this interpretation.

would be red license plates for DUI convicts, which clearly indicates wrongdoing of the vehicle owner. Another helpful example comes from Medicare Five-Star Rating System by the Center for Medicare & Medicaid Services (CMS), which publicizes the highest rank (5 star) and the lowest rank (1 star) and the value increases monotonically with the number of stars.

7 Conclusion

This study evaluates the effectiveness of a novel signal-based policy which changes the visual content of corporate registered vehicles above a price threshold. I document a clear announcement effect where buyers make intertemporal shifts to bring purchase forward. After the implementation date, the policy brings a short term deterrence effect which fades out after 8-10 months. Incorporating anticipation effects leads to a net decline in purchase of 9,401 vehicles. Parameter estimates for each month are consistent with explanations where the relative visibility of the regulated segment decreases over time, as well as the misspecification of shame function.

This study opens up several directions for future research. First, several other margins of tax evasion exist other than new vehicle purchases. Importantly, substitution the used car market, scrappage decisions of the existing fleet and driving patterns of corporate vehicle owners are also of a first-order importance to comprehensively assess the new policy (Gruenspecht, 1982). Second, the fact that we do not directly observe the amount of tax evasion but only through a proxy is an important limitation of our study. While this is a common empirical challenge to the tax compliance literature (Slemrod, 2019), results may also reflect idiosyncractic nature of vehicle purchases rather than pure evasion patterns. Finally, additional research should be conducted in non-Asian context given the importance of non-pecuniary determinants of tax compliance (Luttmer and Singhal, 2014). Previous research documents the importance of cultural norms in firm tax evasion

(Cheng et al., 2024; DeBacker et al., 2015), suggesting that findings should hold conditional on the cultural context. Similar research outside of the East Asian context would further strengthen the external validity of findings.

Appendix A: Supplementary Figures and Tables

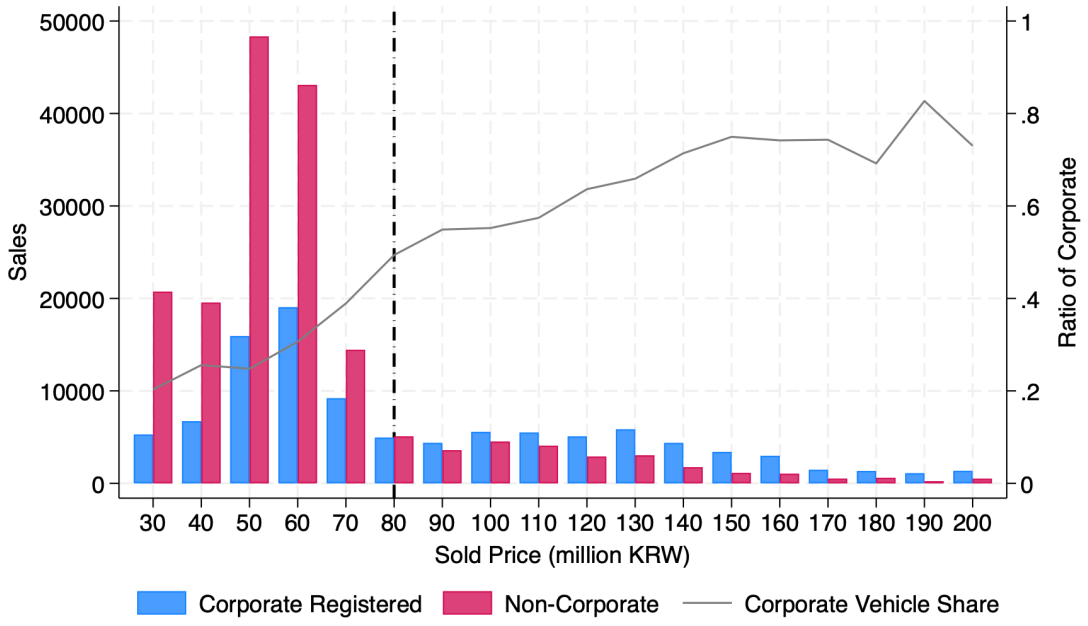

Figure A1: New Sales Count and Corporate Registration per Price Bin, Pre-Policy

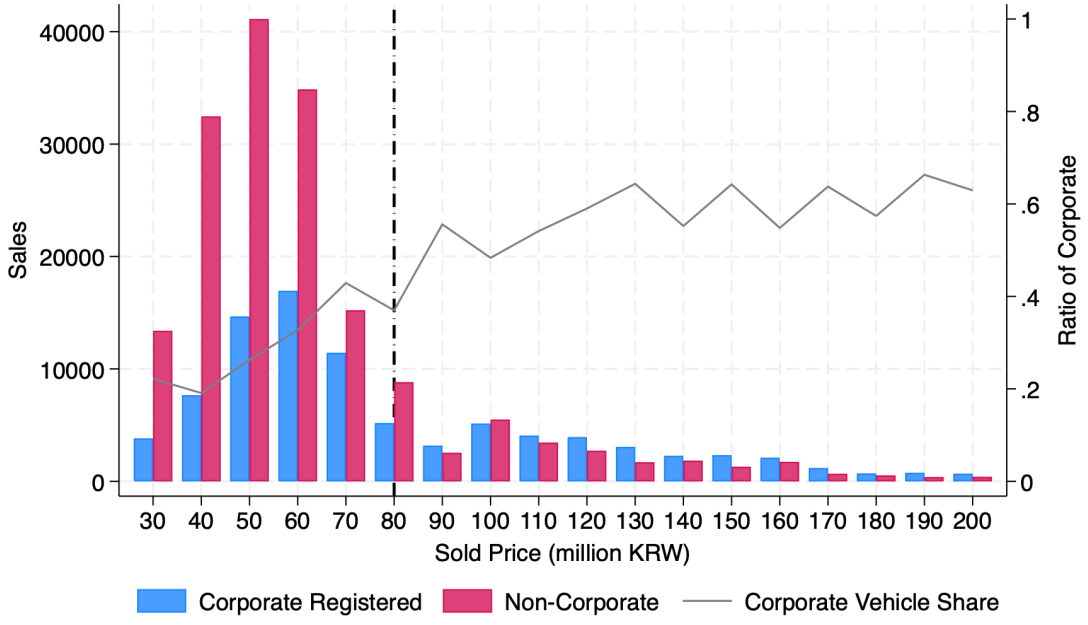

Figure A2: New Sales Count and Corporate Registration per Price Bin, Post-policy

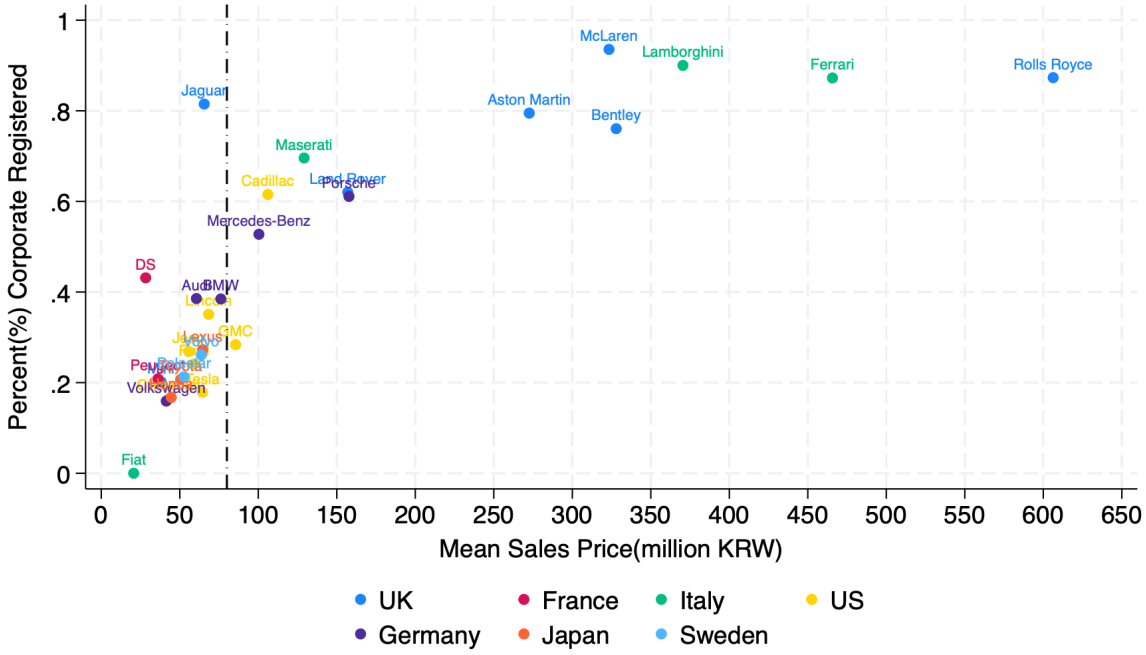

Figure A3: Average Sales Price and Corporate Ratio per Automaker, Pre-Policy

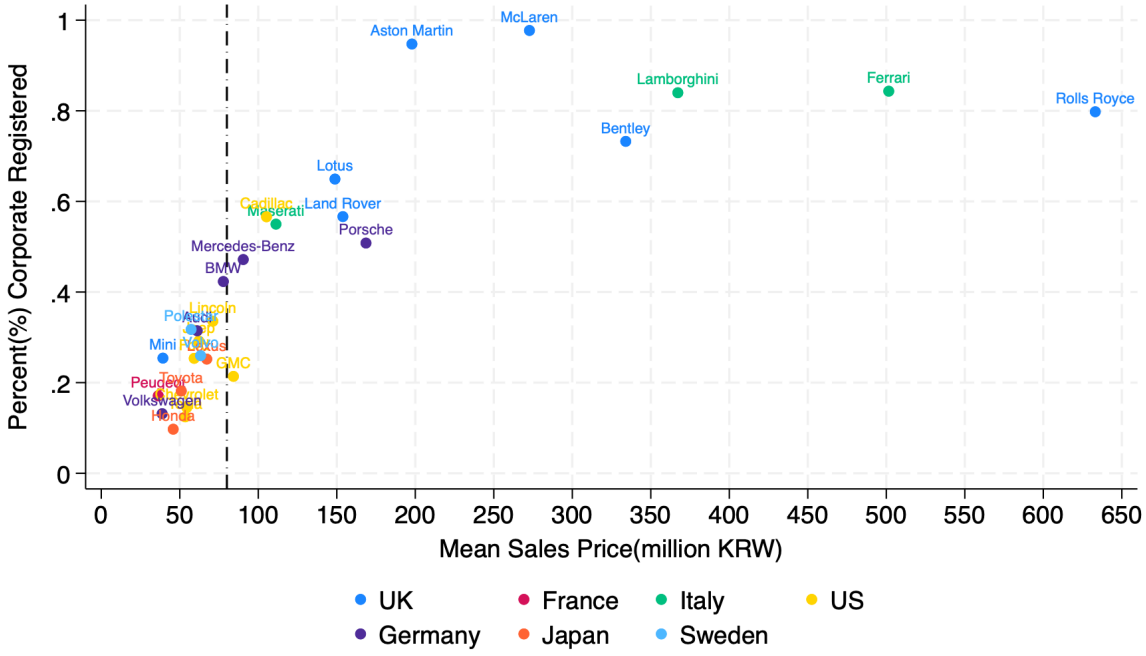

Figure A4: Average Sales Price and Corporate Ratio per Automaker, Post-policy

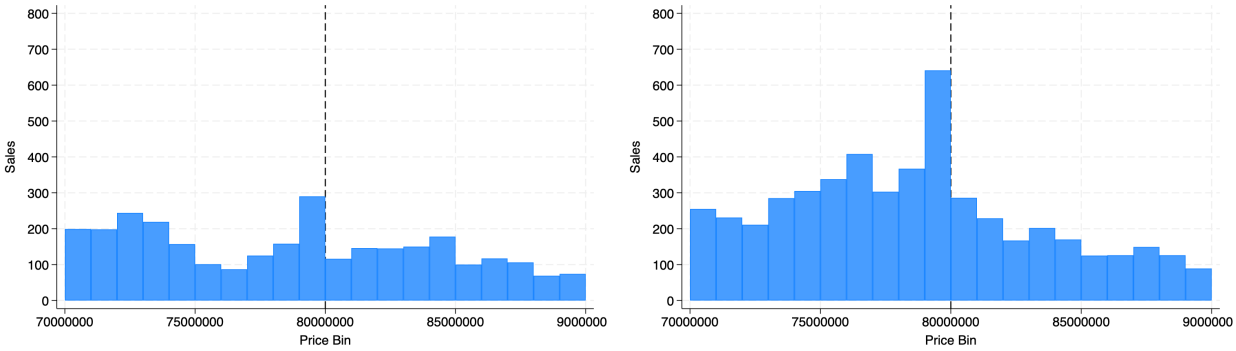

Figure A5: Density of Sales, January-March (left) and April-June (right)

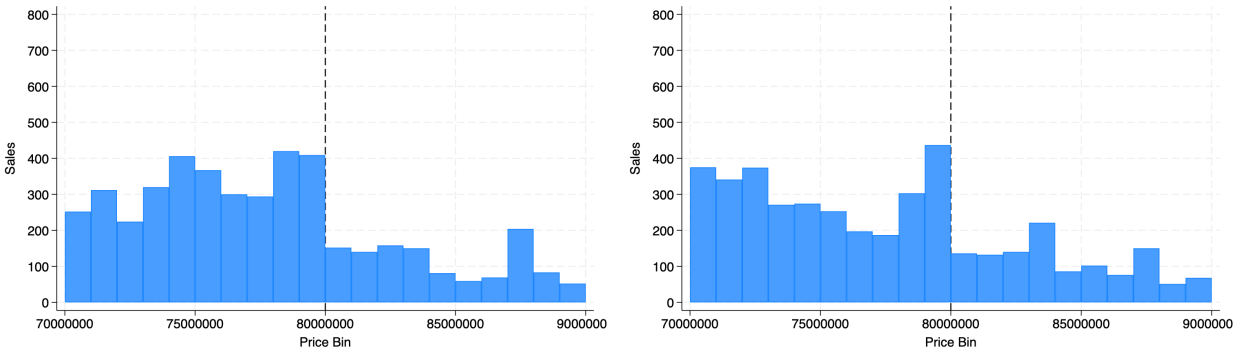

Figure A6: Density of Sales, July-September (left) and October-December (right)

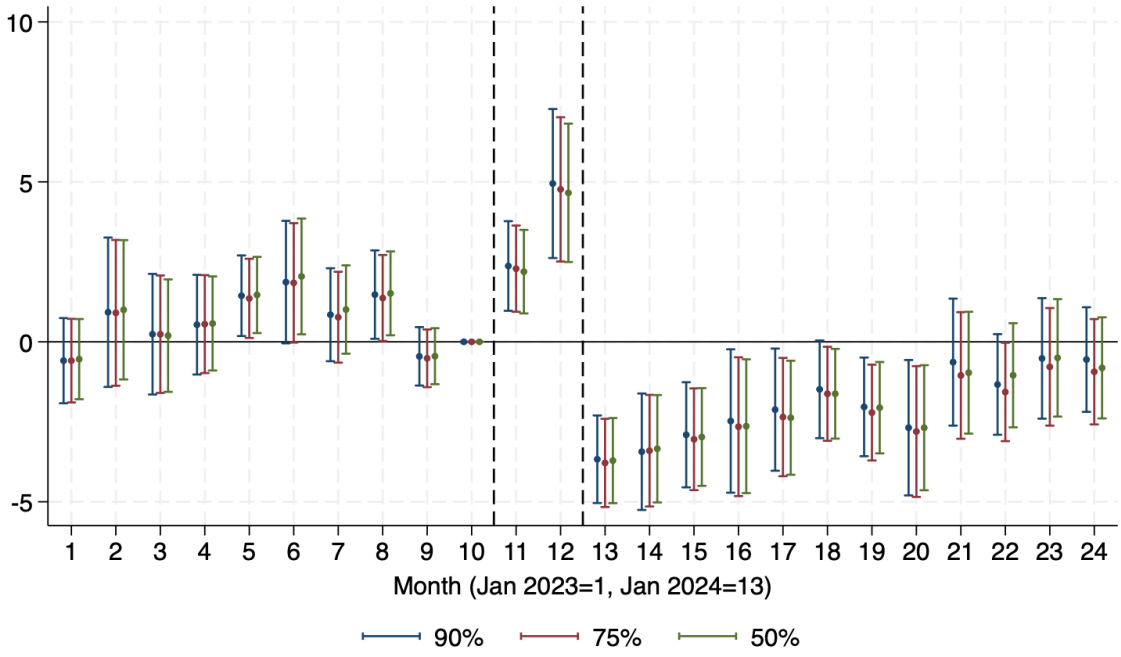

Figure A7: Event Study Estimates per Sample Selection Rule

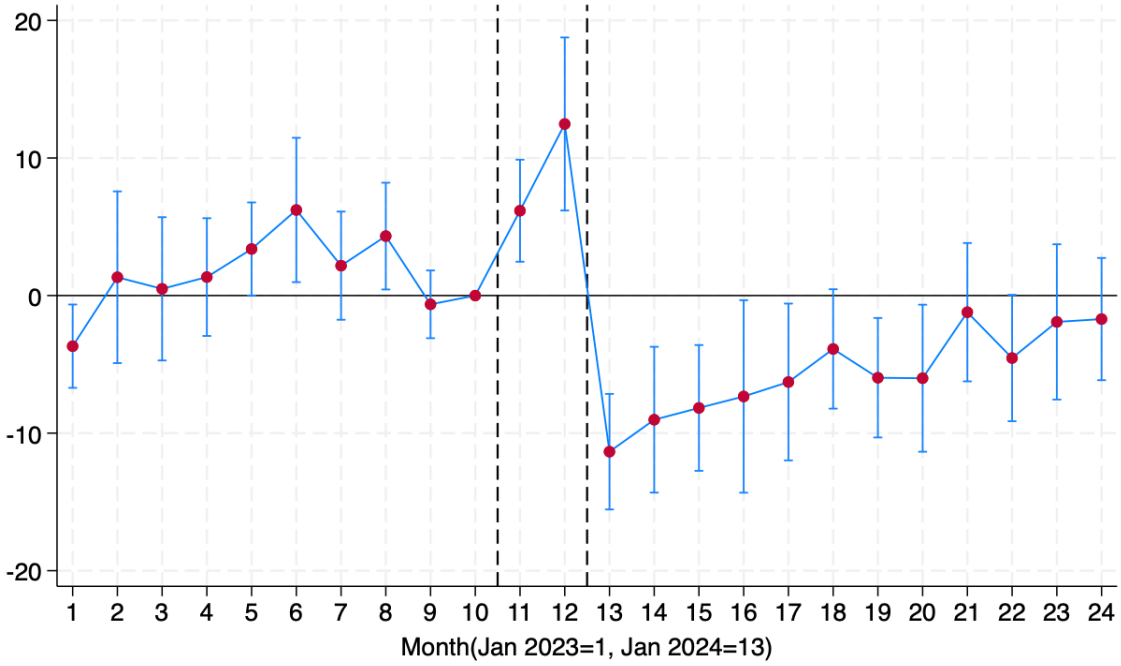

Figure A8: Heterogeneity: Event Study Estimates for German Vehicles

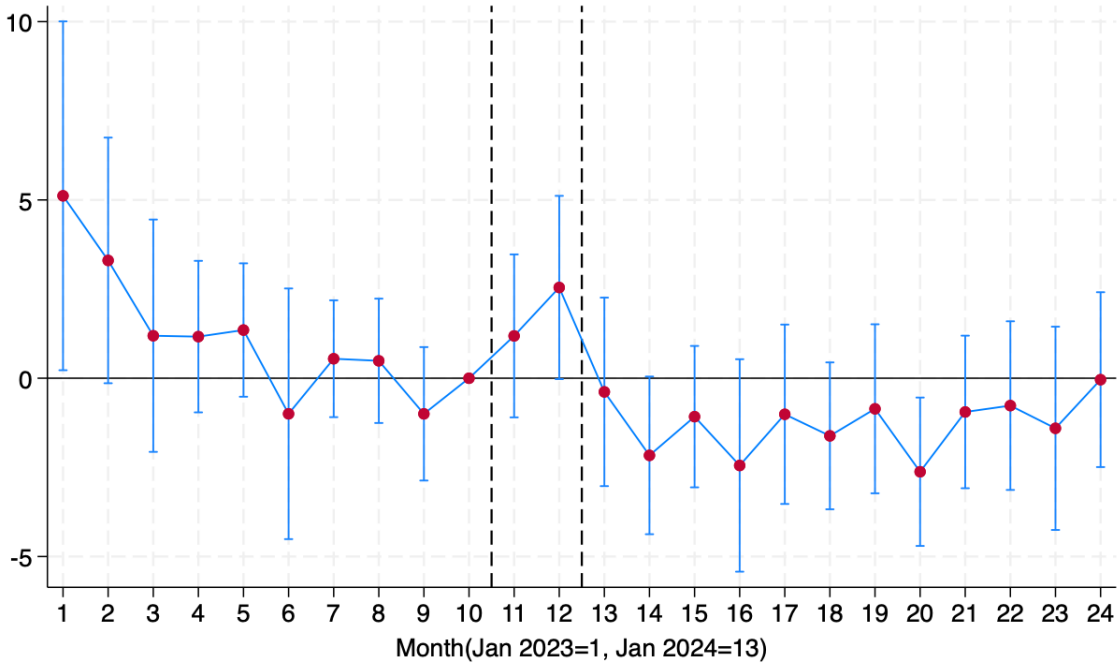

Figure A9: Heterogeneity: Event Study Estimates for Non-German Vehicles

Appendix B: Estimation of Model-level Difference in Differences

This section outlines the model-level approach and numerically compares it to estimates using aggregate sales data, which our approach targets to approximate.

Sample Selection

In order to conduct a model-level analysis, there should be a procedure that selects specific models that sell above the price threshold. I select the sample using the following process:

1. Using pre-policy data only, I calculate the ratio of corporate-registered sales over the price threshold to total corporate sales for each model.
2. **Model Restriction**: I restrict the analysis only to models where the corporate-registered ratio obtained in Step 1 falls above a certain ratio (ex. 10%, 25%, 50%, 75%, 90%)
3. **Sales Restriction**: Within each model, I only include sales where the final transaction price is above the price threshold.

Steps 1 and 2 ensures that we focus on models that are in general sold above the price threshold. Each model would have a predetermined Manufacturer's Suggested Retail Price (MSRP) each month. However, MSRP largely reflects the basic trim without any options attached, whereas in practice a model-month pair could have a range of prices depending on a variety of factors, such as additional options, dealer discounts, customer bargaining sales and so on. Step 3 ensures that within each model, we properly include sales that only occur above 80 million KRW. This owes to the fact that our initial dataset contains transaction-level microdata even though we aggregate saeles at the model-corporate level.

Comparing Estimates

Difference-in-difference estimators (and event study estimators) identify the following average treatment effect on the treated (ATT):

$$\widehat{ATT} = (\bar{Y}_{TRT,Post} - \bar{Y}_{TRT,Pre}) - (\bar{Y}_{CTRL,Post} - \bar{Y}_{CTRL,Pre}) \tag{10}$$

Without loss of generality, consider $\bar{Y}_{TRT,Post}$, the average sales of the treatment group post-treatment. The sample analogue of the aggregate sales data and model-level data would yield:

$$\underbrace{\frac{1}{T_{Post}} \sum_{t \geq Post} \sum_{\substack{\{i \in Corp\} \cap \\ \{Price \geq 80m\}}} Sales_{it}}_{Aggregated} \quad vs \quad \underbrace{\frac{1}{N_{G_T}} \frac{1}{T_{Post}} \sum_{t \geq Post} \sum_{\{g \in Corp\}} Sales_{gt}}_{Model-level\ Aggregation} \tag{11}$$

Two differences emerge. First, there is no $\frac{1}{N_{G_T}}$ pre-multiplied in the first term as there is only one treatment group, namely an aggregate chunk of corporate-registered cars above the price threshold. Second, model-level approach does not require price threshold restrictions (i.e. $\{Price \geq 80m\}$), as we already select vehicles above the price threshold using Steps 1-3.

Formally, the following relationship holds between our model-level approach ($\widehat{ATT}_M$) and aggregate-level counts ($\widehat{ATT}_A$):

$$
\begin{aligned}
\widehat{ATT}_M &= (\bar{Y}_{TRT,Post} - \bar{Y}_{TRT,Pre}) - (\bar{Y}_{CTRL,Post} - \bar{Y}_{CTRL,Pre}) \\
&= \left\{ \frac{1}{N_{G_T}} \frac{1}{T_{Post}} \sum_{t \geq Post} \sum_{\{g \in Corp\}} Sales_{gt} - \frac{1}{N_{G_T}} \frac{1}{T_{Pre}} \sum_{t < Post} \sum_{\{g \in Corp\}} Sales_{gt} \right\} \\
&- \left\{ \frac{1}{N_{G_C}} \frac{1}{T_{Post}} \sum_{t \geq Post} \sum_{\{g \in Corp\}} Sales_{gt} - \frac{1}{N_{G_C}} \frac{1}{T_{Pre}} \sum_{t < Post} \sum_{\{g \in Corp\}} Sales_{gt} \right\} \\
&= \frac{1}{N_{G_T}} \left\{ \frac{1}{T_{Post}} \sum_{t \geq Post} \sum_{\{g \in Corp\}} Sales_{gt} - \frac{1}{T_{Pre}} \sum_{t < Post} \sum_{\{g \in Corp\}} Sales_{gt} \right\} \\
&- \frac{1}{N_{G_C}} \left\{ \frac{1}{T_{Post}} \sum_{t \geq Post} \sum_{\{g \in Corp\}} Sales_{gt} - \frac{1}{T_{Pre}} \sum_{t < Post} \sum_{\{g \in Corp\}} Sales_{gt} \right\} \\
&= \frac{1}{N} \left\{ \left(\frac{1}{T_{Post}} \sum_{t \geq Post} \sum_{\{g \in Corp\}} Sales_{gt} - \frac{1}{T_{Pre}} \sum_{t < Post} \sum_{\{g \in Corp\}} Sales_{gt} \right) \right. \\
&\left. - \left(\frac{1}{T_{Post}} \sum_{t \geq Post} \sum_{\{g \in Corp\}} Sales_{gt} - \frac{1}{T_{Pre}} \sum_{t < Post} \sum_{\{g \in Corp\}} Sales_{gt} \right) \right\} \\
&\approx \frac{1}{N} \widehat{ATT}_A
\end{aligned}
\tag{12}
$$

where the last equality follows from the fact that the number of treated and untreated models are equal in our data. A small discrepancy will arise from two estimates where several models that record corporate-registered sales above the price threshold will not be included during estimation (as per Step 2).

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